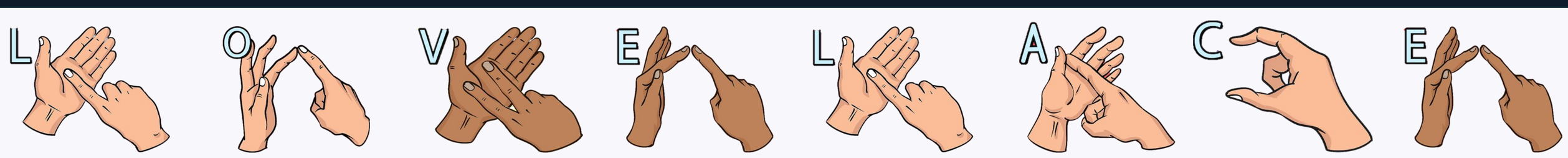
Automatic BSL Transcription

Is BSL transcription feasible under real world constraints?

Project goal: Investigate whether real-time BSL transcription is achievable using computer vision and machine learning with no specialist hardware, making accessible communication available to everyone.



25K

British Sign Language (BSL) is used by **approximately 25,000** people in the UK, yet mainstream video conferencing platforms provide no real-time transcription for signers. This creates a significant barrier to accessible communication in social, educational, and professional environments. For 25,000 BSL users in the UK, **a £0 webcam could replace £3,000+ specialist captioning equipment**.

What I Built

Real-time BSL recognition system running at ~30fps on a standard laptop no specialist hardware and no pre-existing BSL dataset while real time performance is maintained.

Webcam → Landmarks → LSTM → Captions

- Runs in real time on a standard laptop at ~30fps [35 frames] [126 features] [~30fps real-time]
- Currently recognizes 2 BSL signs (expanding to 6 before April)
- 60 training samples per sign, self-collected - no public BSL dataset

LSTM was selected for its ability to capture temporal gesture patterns across sequential frames, whereas CNN alternatives treat each frame independently and miss movement context which is vital for sign language.

ASL vs. BSL

While automatic sign language recognition has been explored for American Sign Language (ASL), BSL remains underrepresented and differs significantly in grammar, vocabulary, and structure. BSL's two-handed alphabet alone doubles the landmark complexity per frame compared to ASL approaches.

FEATURE	BSL	ASL
Alphabet	Two-handed	One-handed
Grammar	Topic-comment structure	Subject-object-verb
Origin	Martha's Vineyard + French SL	French Sign Language
Intelligibility	Not mutually intelligible — separate language systems	
Datasets	Very limited	Well-studied

Prototype Evaluation

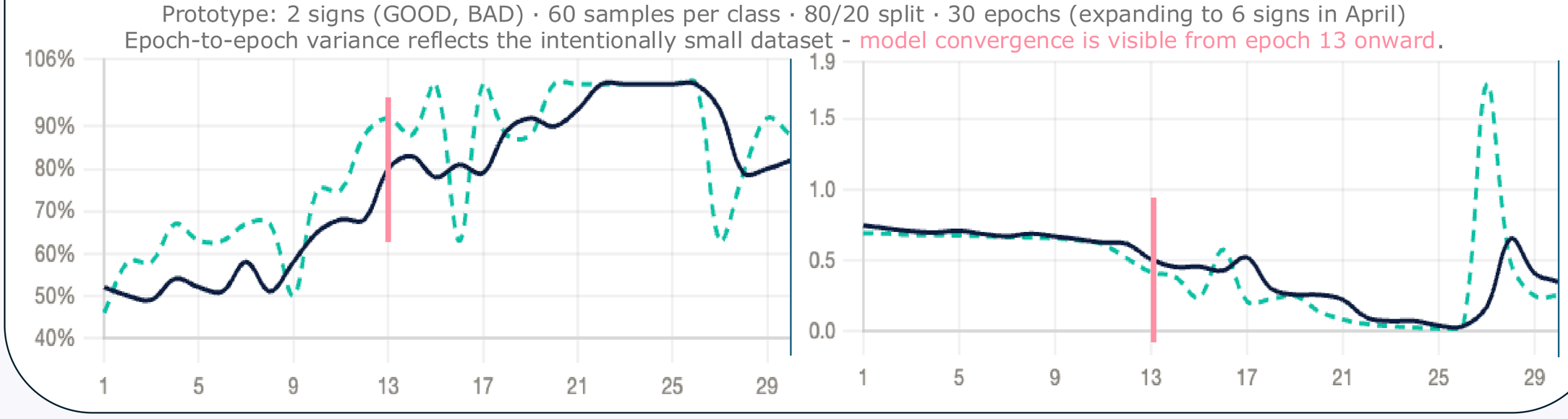
The prototype captures live webcam input, extracts hand landmark sequences, and classifies signs in real time, running entirely on consumer hardware with no specialist equipment. Reliable recognition observed for trained signs during live testing. Scaling to a full BSL vocabulary remains the central challenge, and one this project is actively pursuing. Due to the intentionally small and self-collected dataset, traditional accuracy metrics are not yet representative. Instead, evaluation focuses on real-time performance and system stability.

Future Developments

- Expand dataset to include a larger and more diverse variety of BSL signs
- Incorporate facial and body landmark features
- Improve model robustness across different users
- Extend from isolated signs to continuous sentence recognition

Model Training

Prototype: 2 signs (GOOD, BAD) · 60 samples per class · 80/20 split · 30 epochs (expanding to 6 signs in April)
Epoch-to-epoch variance reflects the intentionally small dataset - model convergence is visible from epoch 13 onward.



SCAN FOR: Demo Video, Source Code and More!

"Built to make everyday communication accessible for the Deaf community" - Grace Carey – Aberystwyth University

Camgoz, N.C. et al. (2018) 'Neural sign language translation', 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition.
Albanie, S. et al. (2021) BBC-Oxford British sign language dataset, arXiv.org. Available at: https://arxiv.org/abs/2111.03635
Koller, O., Forster, J. and Ney, H. (2015) 'Continuous sign language recognition: Towards large vocabulary statistical recognition systems handling multiple signers', Computer Vision and Image Understanding.